Water Jug Problem using Rule-Based Approach (practical no.1)

CODE:

```
jug1_capacity = 4

jug2_capacity = 3

goal = 2

# Initial State

jug1 = 0

jug2 = 0

print("Initial State:", (jug1, jug2))

# Rule 1: Fill Jug1

jug1 = jug1_capacity

print("Fill Jug1:", (jug1, jug2))

# Rule 2: Pour Jug1 -> Jug2

transfer = min(jug1, jug2_capacity - jug2)

jug1 = jug1 - transfer

jug2 = jug2 + transfer

print("Pour Jug1 into Jug2:", (jug1, jug2))

# Rule 3: Empty Jug2

jug2 = 0

print("Empty Jug2:", (jug1, jug2))

# Rule 4: Pour Jug1 -> Jug2

transfer = min(jug1, jug2_capacity - jug2)

jug1 = jug1 - transfer

jug2 = jug2 + transfer

print("Pour Jug1 into Jug2:", (jug1, jug2))

# Rule 5: Fill Jug1
```

```
jug1 = jug1_capacity
print("Fill Jug1:", (jug1, jug2))
# Rule 6: Pour Jug1 -> Jug2
transfer = min(jug1, jug2_capacity - jug2)
jug1 = jug1 - transfer
jug2 = jug2 + transfer
print("Pour Jug1 into Jug2:", (jug1, jug2))
# Goal Test
if jug1 == goal or jug2 == goal:
    print("Goal Reached:", (jug1, jug2))
else:
    print("Goal Not Reached")
```

Output:

